Testing a Willmore-Anisotropy Inequality for 3D Reconstruction and Mesh QA

Pre-registration-ready experimental blueprint and implementation audit

Independent mathematical and computer-vision assessment

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Executive verdict

Scientific status: The proposed strengthened inequality is mathematically meaningful as a bound on **spatial rim length**. For *uniformly averaged visible orthographic silhouettes* it is also a valid, weaker necessary bound. But a self-consistency penalty evaluated on the **same candidate mesh** is identically inactive in the continuum on all admissible meshes, including incorrect reconstructions. Consequently, it cannot inherently be a discriminator of reconstruction quality.

Primary falsifiable question: Does combining externally observed uniformly sampled silhouette perimeter with the curvature terms of a candidate mesh improve reconstruction or reject inconsistent reconstructions beyond a properly tuned direct silhouette loss and standard Willmore regularization? This is empirical, unproved, and vulnerable to noisy curvature estimation.

1. Definitions and mathematical validity

Let $S \subset \mathbb{R}^3$ be a smooth, compact, oriented, embedded, boundaryless surface; $A = \int_S dA$, $W = \int_S H^2 dA$ with $H = (\kappa_1 + \kappa_2)/2$, and χ its Euler characteristic. Define

$$D = \int_S \frac{(\kappa_1^2 - \kappa_2^2)^2}{\kappa_1^2 + \kappa_2^2}\, dA,$$

with zero integrand at zero-curvature points. If L_{rim} is the directional mean of **full spatial rim length**, the audited refinement reads

$$L_{\text{rim}}^2 \le \frac{A}{2}(W - \pi\chi) - \frac{A}{64}D.$$

For a visible orthographic silhouette of an opaque object, let s_{vis} be the uniformly averaged planar perimeter, counting image-set boundary components consistently. Since $s_{\text{vis}} \le L_{\text{rim}}$, one obtains the necessary condition

$$\boxed{W \ge \pi\chi + \frac{2s_{\text{vis}}^2}{A} + \frac{D}{32}.} \tag{1}$$

For **every** smooth admissible surface, equation (1) holds when the measurements come from that same surface. A candidate-only hinge $[\pi\chi + 2s_{\text{pred}}^2/A + D/32 - W]_+$ therefore supplies no theoretical gradient for optimization. This is a **structural null result**, not an implementation flaw.

For a candidate $\widehat{S}$ and independently observed s_{obs}, the residual

$$G(\widehat{S}, s_{\text{obs}}) = \pi\hat{\chi} + \frac{2s_{\text{obs}}^2}{\hat{A}} + \frac{\hat{D}}{32} - \hat{W}$$

can be positive. It then signals an incompatibility of this candidate's intrinsic descriptors with the external silhouette summary **assuming comparable metric calibration, appropriate viewing directions, and reliable estimates**. It does not directly prove the candidate is geometrically wrong.

Failure modes and limits. (i) W can grow due to small-scale wrinkles, so false shapes can easily satisfy the condition; (ii) closed invisible internal cavities cannot be recovered from silhouettes; (iii) genus/chi is unstable for noisy topology; (iv) real cameras typically have limited, nonuniform viewing directions; (v) s^2/A and W, D, χ are invariant under uniform scaling, so the inequality cannot recover absolute scale; (vi) open surfaces, triangle soups, and self-intersecting meshes lie outside the theorem.

Comparison caveat. A convex-hull width/perimeter inequality may be evaluated for connected nonconvex surfaces, but the *hull* discards cavities and concavities. It is not equivalent to the original full-rim inequality.

2. Hypotheses, arms, and fair A/B evaluation

Use exactly the same input images, camera calibrations, initialization, backbone architecture, training steps, image/silhouette losses, mesh resolution, and random seeds for paired runs.

Null hypotheses: H0-R (reconstruction): adding the new penalty does not improve held-out 3D accuracy compared with direct silhouette training. H0-Q (QA): the observation-conditioned residual does not increase detection at a fixed false-rejection rate compared with simple silhouette disagreement, mesh validity checks, and conventional smoothness metrics.

Arm	Losses / score	Purpose
R0	Photometric/depth loss + direct per-view silhouette loss + standard mesh priors	Primary baseline
R1	R0 + tuned conventional Willmore/bending regularizer	Tests regular curvature regularization
R2	R0 + candidate-only hinge using its own predicted s	Negative control; theoretically always zero
R3	R0 + externally observed s_{obs} with $D = 0$	Tests observation-conditioned unrefined inequality
R4	R0 + externally observed s_{obs} with full $D/32$	Tests incremental value of anisotropy

Optional R5: use an independently computed convex-hull mean-width condition as a separate QA baseline, not a replacement for the same-mesh or observation-conditioned arms.

For R4 use an explicitly named experimental loss (it is **not** a standard shape prior):

$$\mathcal{L}_{\text{extra}} = \lambda \left[\max\left(0, \frac{\pi\chi + 2s_{\text{obs}}^2/A + D/32 - W}{W + \epsilon}\right)\right]^2 .$$

This term can perversely reward **increasing** W by adding wrinkles, changing area, or reducing anisotropy; inspect each route. Evaluate an additional gradient-stopped W variant to isolate optimiza-

tion effects. For fixed-topology mesh deformation χ is constant; for neural SDFs, topology changes during remeshing, so apply discrete evaluations at checkpoints rather than assuming differentiability.

Do not backpropagate through Open3D raycasting: it is an evaluation renderer. Train silhouettes with a differentiable renderer such as PyTorch3D, keeping Open3D as an independent test oracle.

Design size. Pilot: 48 shapes (balanced by family), two camera regimes, three paired seeds, arms R0/R1/R3/R4: 1,152 runs. Add R2 to a smaller numerical diagnostic subset. Confirmatory test: approximately 240 held-out shapes, two camera regimes, three paired seeds, four selected arms: 5,760 runs. Report wall time and GPU-hour cost. If budget is tight, reduce shape count via paired power calculations from pilot variance, not via cherry-picking scenes.

Cameras: 4 and 12 training views for sparse and moderate regimes; 32+ held-out, approximately uniform orthographic test views. Repeat separately with perspective cameras to quantify theorem/model mismatch. Rotate view sets across seeds. A limited 180-degree real-camera trajectory is **not** an unbiased spherical mean.

3. Dataset and corruption factorial design

Family	Diagnostic role	Recommended generation
Spheres, ellipsoids	Equality, umbilicity, anisotropy, near-convexity	Analytic parametric surfaces
Rounded long cylinders, capsules	Sharp convex regime; curvature imbalance	Smooth implicit/CAD constructions
Ring tori, genus 1	Saddle regions, silhouette holes and occlusion	Parametric torus with R/r sweep
Genus 2/3+ handlebodies	Topology and narrow necks	Smooth implicit level sets, manifold meshing
Concave bowls and star-shaped indents	Hidden/nonconvex contours	Controlled smooth deformations
Sealed internal voids	Fundamental silhouette non-identifiability	Shells with interior boundary components

Generate references analytically whenever possible. Split by **source shape**, not by mesh tessellation or rendered image: all remeshings/views of a shape belong to the same split. Hold out entire shape families for OOD evaluation. Include clean/high-quality meshes for theorem validation, and intentionally invalid meshes only in the **separate invalid-input gate**.

Public datasets: ABC supplies CAD surface descriptions and geometric ground truth for curvature validation; DTU and Tanks and Temples supply multi-view reconstruction settings; Thingi10K deliberately includes manifold defects and should be labeled as invalid/input-QA stress data rather than naively assumed to satisfy the theorem. Sources: ABC, Tanks and Temples, Thingi10K.

Perturbation grid. At three intensities each: isotropic/anisotropic remeshing, controlled high-frequency wrinkles, vertex noise, spike injection, over-smoothing, concavity filling, adding/removing handles, sealed invisible cavities, silhouette misalignment/calibration error, nonmanifold edges, holes in triangulation, self-intersections. A 'geometrically incorrect but valid' mesh must be distinguished from a 'mesh invalid for the theorem'.

Hidden cavities: In a sealed void, two objects can have identical exterior silhouettes while different geometry, topology, area, and curvature. Measure whether a method *abstains* rather than falsely claiming it has discovered the void. Other modalities (depth, tomography, multiview interior access) are required to resolve it.

4. Metrics, labels, and statistical analysis

Reconstruction primary outcome: mesh-to-reference symmetric surface-sampled Chamfer distance, normalized by object bounding-box diagonal. Secondary: normal consistency, F-score at preregistered geometric tolerances, topological match (χ, connected components, cavity count), watertightness, held-out silhouette IoU/boundary-distance, and runtime/VRAM. Report curvature-energy drift and high-frequency roughness separately to expose gaming the constraint.

QA labels: define ACCEPTABLE using independent 3D ground truth (e.g. preregistered normalised Chamfer and F-score thresholds, no unacceptable topology error); define UNACCEPTABLE using those same independent criteria, not inequality satisfaction. For genuine real-world scans where 3D truth is unavailable, use blind expert ratings as secondary labels only, with inter-rater agreement.

Let *positive* mean an unacceptable candidate that should be rejected. Then

$$\text{FPR} = \frac{\text{acceptable rejected}}{\text{all acceptable}}, \qquad \text{FNR} = \frac{\text{unacceptable accepted}}{\text{all unacceptable}}.$$

Measure AUROC, AUPRC, precision, recall, balanced accuracy, risk-coverage curves with abstention, calibration, and rejection recall **at exactly 1% and 5% FPR**. Thresholds are tuned on validation data then frozen; one does not tune them on held-out tests. Compare to (a) held-out silhouette IoU/mask mismatch, (b) conventional Willmore energy, (c) mesh validity tests, (d) hull-perimeter checks, (e) composite discriminators with and without the new normalized residual.

Inference: use paired differences by *shape*, stratify by shape family and view regime, and cluster bootstrap by source object rather than treating thousands of renderings as independent samples. Report 95% confidence intervals, familywise/multiple-comparison correction, and effect sizes, not just significance. Recommended advancement gate: statistically credible improvement in 3D Chamfer or QA recall at fixed FPR, no unacceptable rise in roughness, and a reasonable runtime cost. This is a prespecified *decision rule*, not an asserted result.

5. Numerical and software QA plan

1. **Validity gate:** Reject theorem evaluation on empty, degenerate, open, nonmanifold, nonorientable, or self-intersecting meshes; flag numerical triangles and extreme aspect ratios. Euler characteristic $V - E + F$ is only meaningful after appropriate mesh validation.
2. **Curvature:** cotangent Laplacian gives vector mean curvature; angle defects give Gaussian curvature. Alternatively, estimate κ_1, κ_2 through multi-scale tangent quadratic fits. Compare both and analytic CAD values.
3. **Anisotropy identity:** one may avoid explicitly resolving signs of principal curvatures by using

$$\frac{(\kappa_1^2 - \kappa_2^2)^2}{\kappa_1^2 + \kappa_2^2} = \frac{16H^2(H^2 - K)}{4H^2 - 2K},$$

   with a zero-curvature convention. Negative estimates of $H^2 - K$ indicate discretization inconsistencies and require flags, not silent certification.
4. **Silhouettes:** uniformly sample S^2 with Fibonacci or scrambled Sobol views; cast first-hit *orthographic* rays into a calibrated grid; extract all visible boundary components from the binary foreground. Ray-tracing with Open3D uses `RaycastingScene.cast_rays`; the result uses infinite hit distance for misses. Camera pixel pitch must be converted to object-length units. Compare image raster resolutions (256/512/1024+) and camera rotations to quantify perimeter bias.
5. **Convergence:** icospheres and parametric tori at three or more mesh resolutions; independently increase view counts and image resolutions. Check scale invariance, rigid-motion invariance, Gauss-Bonnet residual, sphere equality, and torus analytic area/Willmore values.

6. **Uncertainty-aware decision:** bootstrap directions and repeat across rendering resolution and mesh refinements. Obtain a conservative interval $[G_{\text{low}}, G_{\text{high}}]$ including systematic discretization uncertainty. Reject only when $G_{\text{low}} > 0$; pass inequality check only when $G_{\text{high}} \leq 0$; otherwise **ABSTAIN**. Even a conclusive violation means 'descriptors inconsistent', not 'surface geometrically bad'.

Observed smoke test (local Python curvature-only check, no Open3D installed): For unit icospheres subdivided 2/3/4 times, applying the *exact* sphere's $s = 2\pi$ to discrete estimates produced apparent positive residuals $G = 0.36346, 0.09230, 0.02321$. A zero-threshold QA gate falsely rejects all three. Estimated W values converge $12.3452, 12.5106, 12.5524$ toward $4\pi = 12.5664$. This is concrete evidence that numerical tolerance is essential.

The same implementation on tori with major/minor radii $(2, 1)$ gave estimated $W = 22.7092, 22.7718, 22.7876$ as resolution increased, tending to the analytic $4\pi^2/\sqrt{3} \approx 22.7929$. The Python file is syntax-checked and curvature-tested. **The Open3D raycasting functions were not executed in this runtime; their API is grounded in Open3D documentation and must be smoke-tested on a machine with Open3D installed.**

6. Repository structure and usage

Recommended organization:

```
experiments/
  configs/{r0,r1,r2,r3,r4}.yaml
  datasets/{shapes,views,splits}.json
  curvatures.py
  silhouette_renderer.py
  train.py
  qa_filter.py
  analyze.py
  tests/{sphere,torus,mesh_validity,raster_resolution}.py
```

The accompanying standalone file `contour_qa_blueprint.py` includes: cotangent and angle-defect curvature, Euler characteristic, approximate principal curvatures, anisotropy, spherical directions, Open3D orthographic raycasts, raster silhouette perimeter, projected convex-hull perimeter, normalized residual, and conservative abstention for incompatible curvature estimates.

To run after installing `numpy scipy scikit-image open3d`:

```
python contour_qa_blueprint.py candidate.ply --views 64 --resolution 512
```

It is a **research prototype**, not a certified numerical verifier or a differentiable training implementation.

7. Conditions for a positive research claim

The inequality will be **reconstruction-useful** only if R4 beats R0/R1 on independent held-out geometry and does not obtain its benefit from increasing W via artificial wrinkles. It will be **QA-useful** only if the externally conditioned score improves rejection over direct observed-vs-rendered silhouette comparison at fixed 1%/5% false-rejection rates, remains calibrated on fine meshes and noisy masks, and correctly abstains on unverifiable inputs. If candidate-only R2 produces a large positive effect, inspect the implementation first: an exact theorem predicts no continuum signal, and such an effect likely comes from numerical artifacts or implicit added regularization.

Overall verdict before experiments: mathematically interesting; implementation QA plausible; statistically useful external-consistency filter unproven; inherent quality-aware candidate-only regularizer unlikely. Do not equate a tight inequality with a high-information reconstruction loss.

Selected technical references

1. Open3D, RaycastingScene API.
2. PyTorch3D, SoftSilhouetteShader.
3. Koch et al., ABC: A Big CAD Model Dataset for Geometric Deep Learning, CVPR 2019.
4. Tanks and Temples benchmark.
5. Zhou and Jacobson, Thingi10K, 2016.
6. Topping, Sharp diameter estimates via integral geometry, October 2026.
7. Miura, Topping’s diameter conjecture for immersed surfaces, November 2026.